

RAPID COMMUNICATION



Second line advanced or metastatic non-small cell lung cancer treatment after chemo-immunotherapy: a retrospective study

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ABSTRACT

Background: With the shift to immunotherapy (IO) in first-line (1L) treatment for patients with advanced non-small cell lung cancer (aNSCLC), this study aims to describe second-line (2L) treatment patterns and outcomes.

Methods: This retrospective study examined 2L outcomes in patients with stage IIIB-IV, epidermal growth factor receptor (EGFR)/-anaplastic lymphoma kinase (ALK) wild-type aNSCLC who started 1L therapy between 2016 and 2019 and received subsequent 2L therapy. Follow-up continued through November 2020. Subgroups included: 2L IO, 2L non-IO, and IO re-challenge (1L and 2L).

Results: A total of 194 patients initiated 2L therapy; 93 received IO (12 also in 1L), and 101 did not. For IO and non-IO, respectively: overall response rate (ORR) was 28.0% and 25.7%; median follow-up was 23.7 and 20.0 months; median realworld progression-free survival (rwPFS) was 8.4 and 5.1 months ($p = 0.0006$); median overall survival (OS) was 13.5 and 7.7 months ($p = 0.0021$). No deaths occurred in the IO re-challenge cohort after a median follow-up of 35.0 months.

Conclusions: rwPFS and OS were consistent with clinical trials of 2L IO regimens. The few patients who received IO in 1L and 2L had favorable rwPFS and OS. Future research on sequential IO therapies in 1L and 2L aNSCLC is warranted.

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IMMUNOTHERAPY IMPROVES SURVIVAL FOR LUNG CANCER PATIENTS NEEDING SECOND TREATMENT- PLAIN LANGUAGE SUMMARY

Over the past few decades, lung cancer research has greatly advanced our understanding of the disease. We now know that lung cancer is not just one type of cancer, but rather 20 or more unique types, each caused by different genetic mutations. For patients whose genetic tests do not find a specific target for treatment, immunotherapy (IO), often combined with chemotherapy, has become the preferred option.

Our study focuses on understanding the outcomes of these patients outside of clinical trials. We used data from patients' medical records, which were reviewed and abstracted by the doctors who treated them. We looked at patients who received IO or other treatments as their second option after their first treatment stopped working. We found that patients who received IO as their second treatment lived without their cancer getting worse for about 8.4 months, compared to 5.1 months for those who received other treatments. Overall, it was observed that patients who received IO lived for about 13.5 months, while those who received other treatments lived for about 7.7 months. A small group of patients who received IO both as their first and second treatment did not have any deaths during the study period. These findings may help doctors make better treatment decisions for patients who need a second treatment after their first one stops working, but more research is needed to confirm these results.

1. Introduction

Lung cancer remains the leading cause of cancer-related mortality in the United States (U.S.) [1]. The arrival of immunotherapy (IO) has revolutionized the treatment landscape for advanced or metastatic non-small cell lung cancer (aNSCLC) [2].

For aNSCLC patients without targetable mutations, IO therapy with immune checkpoint inhibitor (ICI) therapy plus/minus a cytotoxic T-lymphocyte antigen 4 inhibitor with or without chemotherapy has become the standard of care [3,4]. An observational, electronic medical records analysis of 501

aNSCLC patients who initiated first-line (1L) therapy between 2016 and 2019 found that 25.5% received chemotherapy, 27.1% received IO monotherapy, and 47.3% received IO combination therapy (most commonly IO + platinum doublet chemotherapy [PDC]) [5]. Similarly, results of previous research through a medical chart review demonstrated that among 497 patients who initiated 1L systemic therapy during 2016, the most common 1L regimens were PDC plus IO (PDC + IO) (40.6%), PDC (29.4%), IO monotherapy (20.7%), and PDC + bevacizumab (6.2%) [6].

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Article highlights

- Advanced or metastatic non-small cell lung cancer (aNSCLC) treatment paradigms are evolving.
- With immunotherapy (IO) being a common first-line (1L) treatment, less is known about treatment outcomes involving IO in second line (2L).
- 2L treatment may or may not include IO.
- This study examined 2L outcomes in patients with and without prior 1L IO.
- 194 patients initiated 2L. 93 received 2L IO (12 received IO in 1L and 2L) and 101 did not.
- Median overall survival was 13.5 months for 2L IO and 7.7 months for 2L non-IO.
- The overall response rate was 28.0% for 2L IO and 25.7% for non-IO.
- Median real-world progression-free survival was 3.3 months longer in 2L IO vs. non-IO.

Historically, single-agent chemotherapy (SAC) was the fundamental approach in treatment of second-line (2L) nonsmall cell lung cancer (NSCLC) after progression on initial platinum-based combinations until the adoption of 2L IO monotherapy with the approval of nivolumab and pembrolizumab in 2015 and atezolizumab in 2016. Later, docetaxel and nintedanib received recommendation through the Spanish Delphi consensus statement [7]. Secondline treatment options for patients with aNSCLC following ICI are important because of the ongoing, unmet needs for improved survival and tolerability. However, few real-world studies are available to date to describe outcomes for IO pre-treated patients [8–10], especially in the U.S [8]. Moreover, to date, there are limited data describing outcomes of patients who received IO rechallenge, an approach that holds promise based on limited evidence from prospective trials [11,12].

The aims of this research were to broadly examine real-world outcomes for patients in 2L with and without prior IO treatment in 1L. A subset of patients who received IO in both 1L and 2L (defined as IO-rechallenge) was also identified to provide further evidence on that approaches' effectiveness.

2. Materials & methods

2.1. Study design

We conducted a retrospective, medical chart-review that used data collected from community oncology clinics across the U.S. who are part of the Cardinal Health Oncology Provider Extended Network (OPEN). OPEN includes physicians from private, community oncology clinics across the U.S. irrespective of their electronic medical record system (EMR), group purchasing organization, or other requirements. Participation is on a study-by-study basis. The physicians performed patient selection according to pre-specified inclusion and exclusion criteria and completed electronic case report forms (eCRFs) which collected all relevant demographic, clinical, treatment, and outcomes data.

Physicians were asked to randomly select the patients and could enter up to 20 patients' data (to minimize single site bias). In brief, up to 500 patients who had a diagnosis of aNSCLC, determined ineligible for curative treatment, had epidermal growth factor receptor (EGFR)/anaplastic lymphoma

kinase (ALK) wild-type aNSCLC, and initiated any 1L systemic therapy between 1 January 2016 and 12/31/2019, and were ≥ 18 years of age at initiation, were eligible. Full details of the methods the selection criteria and methods have been previously described [6]. Patients were followed from their initiation of 1L therapy until death, lost to follow-up, or at the end of the study period (November 2020), which included any time and experience while receiving 2L or later lines of therapy (LOTs). The patients of interest included in the analytic dataset for this study were those who initiated 2L regardless of 1L treatment. The Western Institutional Review Board reviewed the study protocol and eCRF and the board approved a request for a waiver of authorization for use and disclosure of protected health information (PHI) for this research.

For statistical analysis, 3 cohorts were identified: 1) Those treated with IO in 2L, 2) those not treated with IO in 2L, and 3) those who were treated with IO in both 1L and 2L (IO rechallenge- a subgroup of cohort 1). Demographic and clinical characteristics of the cohorts' outcomes measured included disease response calculated from reported tumor lesion measurements abstracted by physicians at initiation of both 1L and 2L and then at scans while on treatment as previously described [6,13]. Physicians also recorded the time to disease progression as confirmed either through radiographic studies or by clinical judgment. Median realworld progression-free survival (rwPFS, i.e., the middle point of time at which patients, in a real-world setting, live without their cancer progressing after treatment initiation) in 2L was defined as the interval from initiation of the 2L therapy until disease progression or death due to any cause. Patients who had not yet experienced disease progression were censored on their last visit date at the oncologist's clinic. Median overall survival (OS, i.e., the middle point of time at which half of the patients are still alive and half have died, starting from the initiation of treatment and including deaths from any cause) was defined as the interval from initiation of 2L therapy until death. Patients alive at the end of the follow-up period were censored on their last reported visit date at the practice. rwPFS and OS were estimated by using the Kaplan–Meier method. The 2L IO and 2L non-IO cohorts were summarized descriptively in terms of the patient characteristics and clinical outcomes including tumor response to 2L, rwPFS in 2L, and OS from initiation of 2L. Kaplan–Meier analyses were performed to describe rwPFS and OS. The IO rechallenge cohort was described independently.

3. Results

3.1. Physician abstractors

A total of 46 physicians participated in the study. A majority (87%) practiced in the community setting, 8.6% in an academic medical center/teaching hospital and 4.4% reporting "other." Nearly half (47.8%) of oncologists practiced in urban settings, and the remainder were in suburban (34.8%) and rural (17.4%) settings. All 4 U.S. regions were represented, with the West representing the greatest proportion of oncologists (34.8%). Participating oncologists had a median 16 years in practice and managed a median of 41 aNSCLC patients in the year prior to data collection.

3.2. Treatment patterns

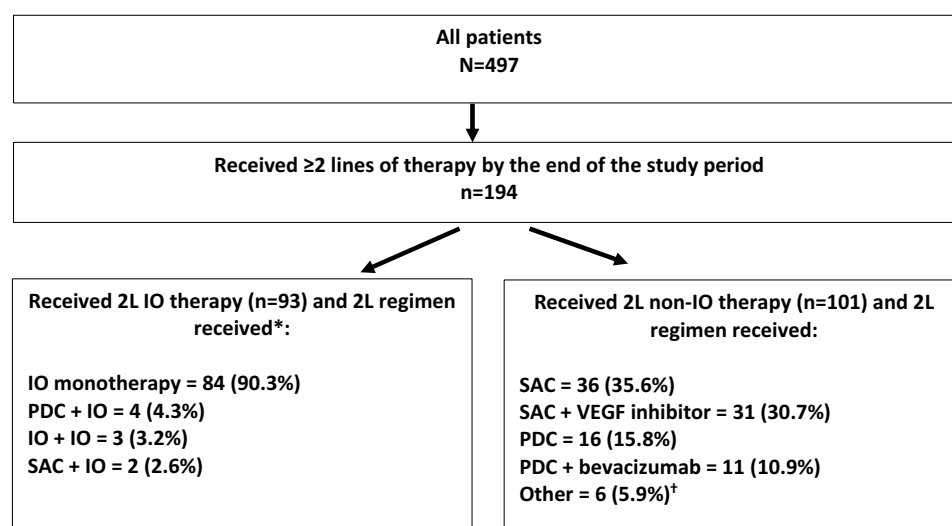
Of the total 497 1L patients included in the analytical database submitted by 46 physicians, 194 received ≥ 2 LOTs. Of those, 194 who received a 2L, 93 were treated in 2L with an IO regimen and 101 were treated in 2L with a non-IO regimen (Figure 1). The median follow-up time from the start of 2L was 18.8 months for all patients who received 2L treatment, 23.7 months for those who received 2L IO, and 20.0 months for those who received 2L non-IO. Of these 194, 12 patients were identified who received IO in both 1L and 2L and were included in the IO rechallenge cohort. Among 93 treated in 2L with IO, 2L treatments consisted of IO mono in 90.3%, PDC + IO in 4.3%, IO + IO in 3.2%, and SAC + IO in 2.6% (Figure 1). Among 101 treated in 2L with non-IO, 2L treatments consisted of SAC in 35.6%, SAC + vascular endothelial growth factor (VEGF) inhibitor in 30.7%, PDC in 15.8%, PDC + bevacizumab in 10.9%, other in 5.9% and non-PDC in 1.0% (Figure 1). The majority of 2L IO treated patients initiated 1L therapy in 2016 and 2017 (81.7%) and for the 2L non-IO treated patients fewer initiated 1L therapy in 2016 (12.9% vs. 41.9% in the 2L IO treated cohort) while 39.6% initiated 1L 2L non-IO in 2017, 24.8% in 2018, and 22.8% in 2019. Treatment patterns in 1L among these cohorts have been previously described but overall, in the 2L IO treated, 65.9% received 1L PDC and 18.7% received 1L PDC + bevacizumab whereas 61.4% of the 2L non-IO treated received 1L PDC + IO and 20.8% received 1L IO monotherapy. A detailed listing of treatment regimens received in 1L and 2L in each cohort is provided in the Supplemental Material Table S1-S4 and a visual representation of the treatment regimens is presented with a Sankey diagram in Figure 2.

3.3. Patient characteristics

Demographic and clinical characteristics are shown in Table 1. Across cohorts, the majority of patients were male (52.7% 2L IO; 58.4% 2L non-IO) and non-Hispanic white (61.3% 2L IO; 72.3% 2L non-IO). Median age at initiation of 1L was 65 (range 45-86) in the 2L IO cohort and 62 (range 41-90) in the 2L non-IO cohort. In both cohorts the majority of patients were stage IVB at initial diagnosis (68.8% 2L IO; 62.4% 2L non-IO), had non-squamous histology (62.4% 2L IO; 73.3% 2L non-IO), and had Eastern Cooperative Oncology Group Performance Status (ECOG-PS) 0/1 at 1L initiation (76.3% 2L IO; 79.2% 2L non-IO). The mean Charlson Comorbidity Index (CCI) was 3.5 (standard deviation [sd] = 1.6) in the 2L IO cohort and 3.0 in the 2L non-IO cohort (sd = 1.1). The percentages of Programmed Death Ligand (PD-L1) testing expression levels were unknown for 17.2% of 2L IO treated and 6.9% of 2L non-IO treated patients, and 5.4% of 2L IO treated and 22.8% of 2L non-IO treated patients had $\geq 50\%$ PD-L1 expression.

3.4. Clinical outcomes

Tumor response to 2L therapy in the cohorts was assessed based on the tumor lesion reports and retrospectively guided by Response Evaluation Criteria in Solid Tumors (RECIST) v1.1 guidelines regarding the degree of tumor shrinkage (Table 2). Although the guidelines were developed for prospective measures of response [14], the current study used RECIST v1.1 as a guide to assess retrospective data. The Overall Response Rate (ORR) in 2L for the IO-treated cohort was 28.0% and 25.7% in the 2L non-IO treated. The ORR for 1L treatment was 43.0% in the 2L IO cohort and 58.4% in the 2L non-IO cohort. The duration of 2L disease response was a mean of

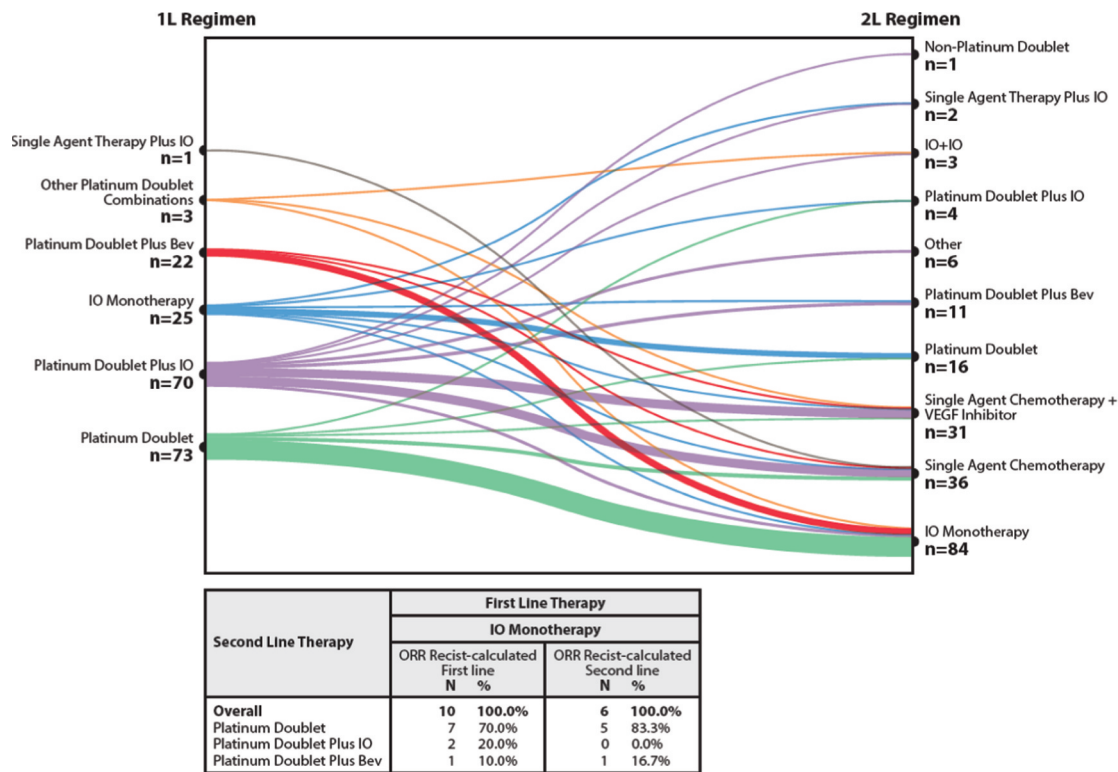


1L = first-line; 2L = second-line; IO = immunotherapy; PDC = platinum doublet chemotherapy; SAC = single-agent chemotherapy; VEGF = vascular endothelial growth factor.

† 5 patients received ramucirumab and 1 patient received bevacizumab.

* A subgroup of the "received 2L IO therapy" cohort is the n=12 patients (IO rechallenge subgroup) who received IO in both 1L and 2L.

Figure 1. Study analysis cohorts and treatment patterns in 2L.



1L = first-line; 2L = second-line; IO = immunotherapy; ORR = overall response rate; VEGF = vascular endothelial growth factor.

Figure 2. Sankey diagram of 1L and 2L treatment paths.

Table 1. Demographic and clinical characteristics of the study cohorts.

	All patients n = 194		2L IO n = 93		2L non-IO n = 101		2L IO Rechallenge n = 12	
Sex at Birth, Male, n (%)	108	(55.7)	49	(52.7)	59	(58.4)	4	(33.3)
Patient Race/Ethnicity, n (%)								
White	130	(67.0)	57	(61.3)	73	(72.3)	6	(50.0)
Asian	11	(5.7)	5	(5.4)	6	(5.9)	2	(16.7)
Black/African American	45	(23.2)	29	(31.2)	16	(15.8)	3	(25.0)
Other	8	(4.1)	2	(2.2)	6	(5.9)	1	(8.3)
Patient Age (years)								
Median (min, max)	64	35–91	65	45–86	62	41–90	63.5	45–80
Stage at Diagnosis, n (%)								
Stage IIIB	8	(4.1)	6	(6.5)	2	(2.0)	3	(25.0)
Stage IIIC	7	(3.6)	2	(2.2)	5	(5.0)	1	(8.3)
Stage IVA	52	(26.8)	21	(22.6)	31	(30.7)	2	(16.7)
Stage IVB	127	(65.5)	64	(68.8)	63	(62.4)	6	(50.0)
Histology, n (%)								
Squamous cell	56	(28.9)	32	(34.4)	24	(23.8)	3	(25.0)
Mixed histology	6	(3.1)	3	(3.2)	3	(3.0)	1	(8.3)
Non-squamous	132	(68.0)	58	(62.4)	74	(73.3)	8	(66.7)
ECOG-PS at 1L, n (%)								
0/1	151	(77.8)	71	(76.3)	80	(79.2)	12	(100.0)
≥2	42	(21.6)	22	(23.7)	20	(19.8)	0	(0.0)
Unknown	1	(0.5)	0	(0.0)	1	(1.0)	0	(0.0)
CCI								
mean, (SD)	3.2	(1.4)	3.5	(1.6)	3.0	(1.1)	3.8	(1.7)
PD-1/PD-L1, n (%)								
Unknown	23	(11.9)	16	(17.2)	7	(6.9)	0	(0.0)
<1%	36	(18.6)	19	(20.4)	17	(16.8)	1	(8.3)
1–49%	107	(55.2)	53	(57.0)	54	(53.5)	8	(66.7)
≥50%	28	(14.4)	5	(5.4)	23	(22.8)	3	(25.0)

Key: 1L = first-line; 2L = second-line; CCI = Charlson Comorbidity Index; ECOG-PS = Eastern Cooperative Oncology Group performance status; IO = immunotherapy; SD = stable disease.

Table 2. Disease response in 1L and 2L among patients treated and not treated with IO in 2L.

	All patients n = 194		2L IO n = 93		2L non-IO n = 101		2L IO Rechallenge n = 12	
2L Disease Response, n (%)								
ORR	52	(26.8)	26	(28.0)	26	(25.7)	1	(8.3)
CR	9	(4.6)	7	(7.5)	2	(2.0)	0	(0.0)
PR	43	(22.2)	19	(20.4)	24	(23.8)	1	(8.3)
SD	47	(24.2)	28	(30.1)	19	(18.8)	2	(16.7)
PD	56	(28.9)	26	(28.0)	30	(29.7)	5	(41.7)
Unknown/Too early to tell	39	(20.1)	13	(14.0)	26	(25.7)	4	(33.3)

Key: 1L = first-line; 2L = second-line; CR = complete response; IO = immunotherapy; ORR = overall response rate (sum of CR + PR); PD = progressive disease; PR = partial response; SD = stable disease.

13.2 months (sd = 10.8) and 6.5 months (sd = 5.5) among the 2L IO treated and 2L non-IO treated cohorts, respectively.

At the time of data collection, 38.1%, 37.6%, and 38.6% of patients were alive among all patients who received 2L treatment, those treated in 2L with IO, and those treated in 2L with non-IO, respectively. Among 92 patients who were treated with 2L IO, median rwpFS was 8.4 months (95% confidence interval [CI]: 6.0–13.2) and 5.1 months (95% CI: 3.9–6.0) for patients who did not receive IO therapy in 2L ($p=0.0006$; Figure 3(a)). Similarly, OS from initiation of 2L was 13.5 months (95% CI: 9.8–18.1) for those patients who received 2L IO therapy and 7.7 months (95% CI: 6.0–9.7) for those who did not receive 2L IO ($p=0.0021$; Figure 3(b)).

3.5. IO rechallenge subgroup

Overall, there were 12 patients identified who had received IO in both 1L and 2L. There were 8 females and 6 were non-Hispanic white, with a median of 63.5 years of age (range 45–80) at initiation of 1L. In terms of clinical characteristics, 6 patients had stage IVB disease at initiation of 1L, eight patients had non-squamous histology, all were ECOG-PS 0/1, PD-L1 expression levels were: <1% for 1 patient, 1–49% for 8 patients, and $\geq 50\%$ for 3 patients. Tumor response rates in 1L were 25.0% (with 8.3% or 1 patient unknown) and 8.3% in 2L (with 33.3% unknown). Median rwpFS was estimated at 30.9 months (95% CI: 7.5–not reached) in 2L. No deaths were

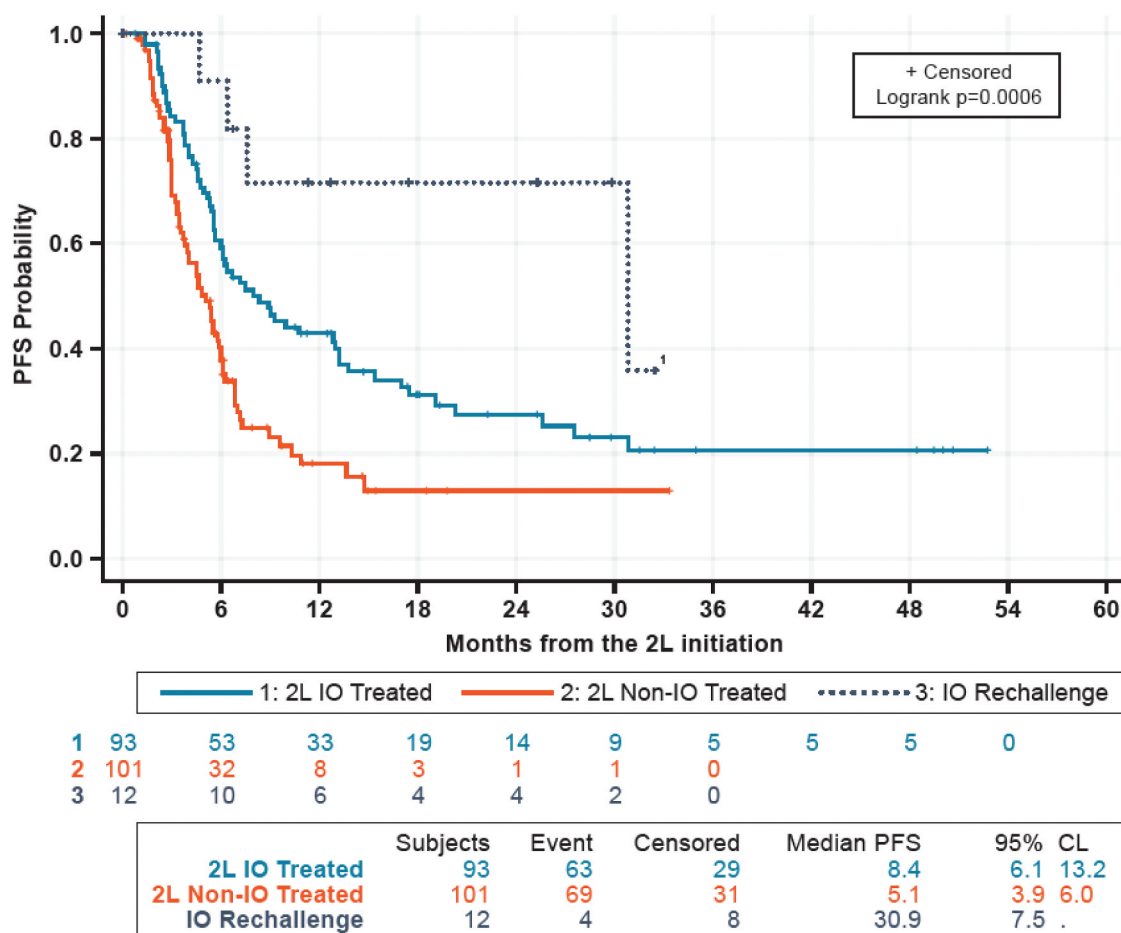


Figure 3. Real-world progression-free survival (rwpfs) in 2L io-treated, non-io-treated, and io rechallenge patients.¹

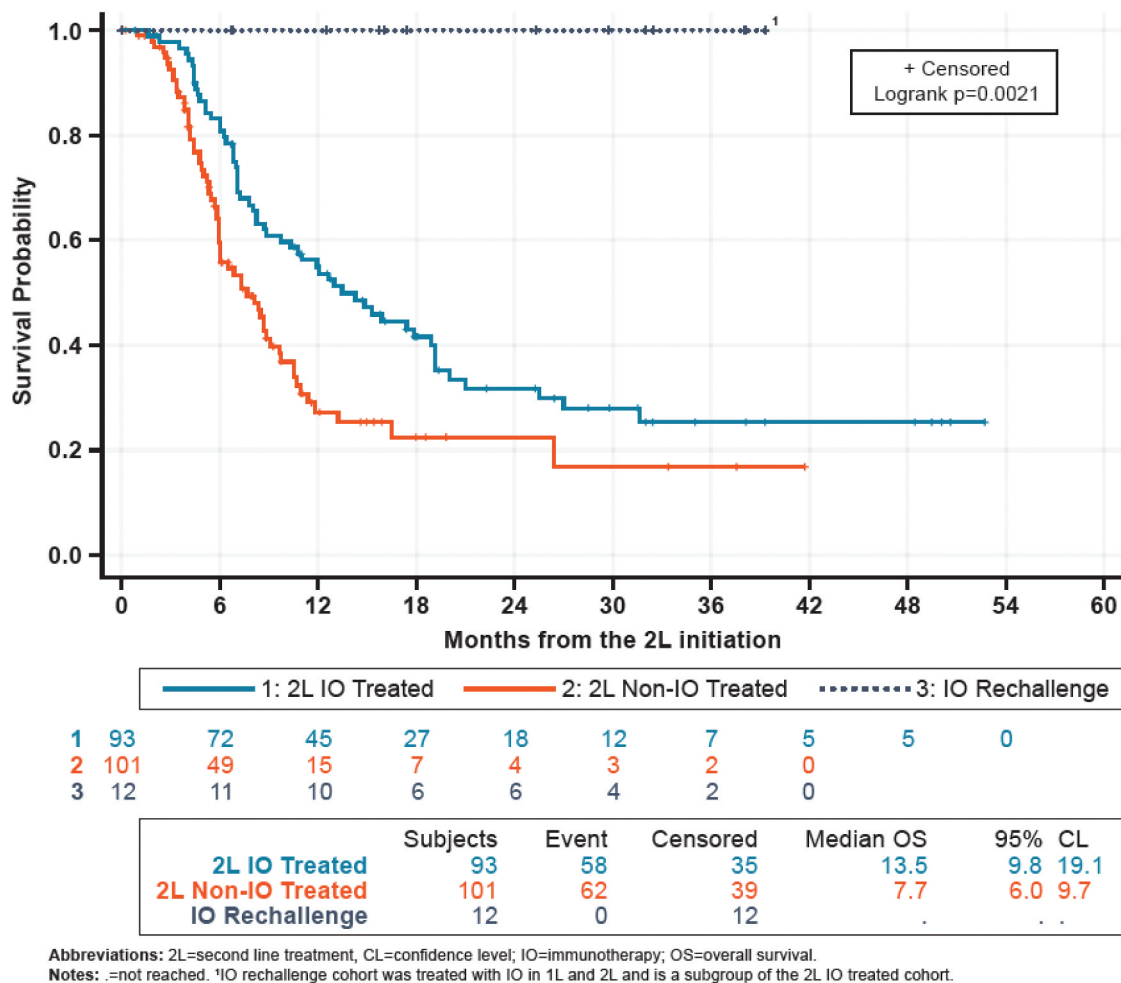


Figure 3. (Continued.)

reported in this cohort and median follow-up for these patients was 35.0 months (range 10.4–47.6).

4. Discussion

Immunotherapy has disrupted the 1L treatment paradigm and has improved outcomes for patients with a variety of cancers including aNSCLC. IO with or without chemotherapy has become the mainstay for 1L therapy for patients with aNSCLC without targetable oncogenic driver mutations. For some patients with aNSCLC, long-term survival of 5 or 10 years post treatment of IO therapies may become the norm [15]. A recent editorial by Moliner et al. (2023) reiterates the lack of clarity in treatment selection at progression after first-line chemo-immunotherapy [16]. There remains an unmet need for novel treatment paradigms to be explored, and real-world data is one mechanism to shed light and generate hypothesis about the best treatment paradigm for patients. Given the paucity of data in 2L NSCLC following chemo-immunotherapy, this study provides evidence of the outcomes for patients after initiating 2L following IO and non-IO therapy in 1L.

First, patients in this study who received 2L non-IO therapy had a response rate of 25.7%, a rwPFS in 2L of 5.1 months, and an OS from initiation of 2L of 7.7 months. Approximately one-quarter of patients in our study received single agent docetaxel (23.8%) or

docetaxel plus ramucirumab combination (25.7%) in 2L in this cohort. Our results are not directly comparable to prior clinical trials given the heterogeneity of treatments received in 2L for the non-IO cohort. The phase 3 REVEL trial of docetaxel plus ramucirumab found a median progression-free survival (PFS) of 4.5 months, and the LUME-Lung1 study of docetaxel plus nintedanib reported a median PFS of 4.4 months [17], both PFSs similar to that observed in this 2L real-world cohort [18,19]. Clinical trials from the early 2000s of docetaxel demonstrated median PFS of 4 or less months [20]. Prior real-world data have reported median rwPFS of 2.9 months and a median OS of 8.1 months in 2L chemotherapy [21].

Over 90% of patients in the 2L IO cohort received either nivolumab (64.5%), pembrolizumab (12.9%), or atezolizumab (11.8%). Moreover, as previously reported, median rwPFS for 1L in this cohort of patients who predominantly received PDC was slightly higher at 5.3 months than found in other retrospective trials of PDC [6]. One recent, Italian real-world comparison of these 3 treatments found 2L OS of 8.9, 9.4, and 8.7 months, respectively [22]. In this IO rechallenged cohort, no deaths were observed over a 35-month follow-up period, which may explain some of the larger benefit seen.

The findings of the IO-rechallenged cohort were of interest given that no deaths were observed over a fairly long median follow-up time of 35 months. Combinations of

platinum-doublets and IO have been the standard 1L treatment of treatment-naïve aNSCLC over the last 5 years [16]. However, given the likelihood of eventual progression after 1L treatment, a better understanding of 2L treatment outcomes is needed in aNSCLC. In particular, the paucity of data in 2L IO treatment following 1L IO treatment warrants future research to examine IO rechallenge outcomes. Other studies of IO rechallenge treatment in aNSCLC patients have demonstrated clinical benefit [23,24]. The results from our real-world exploration of this treatment approach need further retrospective and prospective study in larger cohorts. Currently, a phase 2 trial of rechallenge with pembrolizumab is ongoing (NCT03526887).

4.1. Limitations

While the results of this research provide new evidence regarding outcomes for patients in 2L in the post PDC + IO 1L era, there are several limitations that should be considered. First, this study may not have been representative of all patients diagnosed with aNSCLC or representative of all physicians treating these patients as a dataset of all patients from each site was not available. Second, the heterogeneity and relatively small sample size in cohorts and subgroups prohibits general extrapolation to the larger aNSCLC patient population. Third, the 2L treatment patterns and outcomes were likely influenced by the therapy received in 1L and this likely confounder should be considered when interpreting outcomes among 2L IO treated and 2L non-IO-treated cohorts. Fourth, 2L insights for specific regimens are limited and may be explored in future studies. A comparison of specific 2L treatments by regimen would require a separate 2L efficacy study across the listed regimens with a stated objective. Finally, even indirect comparisons of the outcomes between groups should be approached with caution as the study was not designed nor powered to detect these differences.

5. Conclusions

As clinical trial populations are often healthier, younger, less diverse and more economically advantaged than patients treated outside trials, trials are often perceived as not representative of the patients seen in routine community practice. rwPFS and OS observed in this real-world patient population are consistent with PFS and OS reported in recent clinical trials of 2L IO regimens and further support its effectiveness. Clinical outcomes observed in real-world patients can help reinforce the benefit of 2L treatment in appropriate patients and advance evidence-based medicine. Finally, the observations in the small number of patients who received IO in both 1L and 2L are provocative and warrants further investigation.

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Disclosure statement

Lyudmila Bazhenova reports a relationship with Janssen Pharmaceuticals Inc that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with Boehringer Ingelheim Corp USA that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with Bristol Myers Squibb Co that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with Merck & Co Inc that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with Regeneron Pharmaceuticals Inc that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with Genentech that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with Daiichi Sankyo Inc that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with Blueprint Medicines Corporation that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with Takeda Pharmaceutical Company Limited that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with AstraZeneca that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with Bayer AG that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with BeyondSpring Inc. that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with G1 Therapeutics Inc that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with AbbVie Inc that includes: consulting or advisory; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with Neuvogen that includes: consulting or advisory and equity or stocks; payments for advisory boards and no speaker fees. Lyudmila Bazhenova reports a relationship with ORIC Pharmaceuticals that includes: consulting or advisory and equity or stocks; payments for advisory boards and no speaker fees. Beilei Cai reports a relationship with Novartis Pharmaceuticals Corporation that includes: employment and equity or stocks. Danielle Gentile reports a relationship with Cardinal Health that includes: employment. Jonathan Kish reports a relationship with Cardinal Health that includes: employment. Bruce Feinberg reports a relationship with Cardinal Health that includes: employment and equity or stocks.

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Author contributions

Lyudmila Bazhenova: Conceptualization, Methodology, Writing – Review & Editing. Beilei Cai: Conceptualization, Methodology, Writing – Review & Editing, Funding acquisition. Danielle Gentile: Conceptualization, Methodology, Validation, Writing – Original Draft, Writing – Review & Editing, Visualization. Jonathan Kish: Conceptualization, Methodology, Writing – Review & Editing. Gabor Kari: Writing – Review & Editing. Bruce Feinberg: Conceptualization, Methodology, Writing – Review & Editing.

Ethical disclosure

The Western Institutional Review Board reviewed the study protocol and the case report form, and the board approved a request for a waiver of authorization for use and disclosure of protected health information for this research. The study is exempt under 45 CFR § 46.104(d)(4). WIRB Work Order #1-1341879-1. Research data were collected through electronic case report forms (eCRFs) from community oncology clinics across the U.S. who are part of the Cardinal Health Oncology Provider Extended Network (OPEN).

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Data sharing statement

Due to the nature of the research, supporting data is not available.

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References

Papers of special note have been highlighted as either of interest (*) or of considerable interest () to readers.**

- Siegel RL, Miller KD, Fuchs HE, et al. Cancer statistics, 2022. Surveillance, epidemiology, and end results program (SEER) cancer stat facts: lung and bronchus cancer. *CA Cancer J Clin*. 2021;71(1):7–33. doi: [10.3322/caac.21654](https://doi.org/10.3322/caac.21654)
- Shields MD, Marin-Acevedo JA, Pellini B. Immunotherapy for advanced non-small cell lung cancer: a decade of progress. *Am Soc Clin Oncol Educ Book*. 2021;41:e105–e127. doi: [10.1200/edbk_321483](https://doi.org/10.1200/edbk_321483)
- This reference highlights a historical perspective of how treatment for advanced non-small cell lung cancer has evolved.**
- Assi HI, Kamphorst AO, Moukalled NM, et al. Immune checkpoint inhibitors in advanced non-small cell lung cancer. *Cancer*. 2018;124(2):248–261. doi: [10.1002/cncr.31105](https://doi.org/10.1002/cncr.31105)
- This reference highlights that in an era of increasing costs of care and potential for toxicities related to immune checkpoint inhibitors, proper patient selection is critical for optimized treatment.**
- Chiang AC, Herbst RS. Frontline immunotherapy for NSCLC — the tale of the tail. *Nat Rev Clin Oncol*. 2020;17(2):73–74. doi: [10.1038/s41571-019-0317-y](https://doi.org/10.1038/s41571-019-0317-y)
- This article describes why the predictive value of PD-L1 and tumour mutational burden remains to be determined.**
- Zhang X, Declue RW, Herms L, et al. Real-world treatment patterns and outcomes in PD-L1-Positive non-Small cell lung cancer. *Immunotherapy*. 2021;13(18):1521–1533. doi: [10.2217/imt-2021-0145](https://doi.org/10.2217/imt-2021-0145)
- Bazhenova L, Kish J, Cai B, et al. Real-world observational study of current treatment patterns and outcomes in recurrent or locally advanced/metastatic non-small cell lung cancer. *Cancer Treat Res Commun*. 2022;33:100637. doi: [10.1016/j.ctarc.2022.100637](https://doi.org/10.1016/j.ctarc.2022.100637)
- This article is a predecessor to the current article focused on treatment patterns and outcomes in the same patient population.**
- Isla D, De Castro J, García-Campelo R, et al. Treatment options beyond immunotherapy in patients with wild-type lung adenocarcinoma: a delphi consensus. *Clinical & translational oncology: official publication of the federation of spanish oncology societies and of the national cancer institute of mexico*. 2020;22(5):759–771. doi: [10.1007/s12094-019-02191-y](https://doi.org/10.1007/s12094-019-02191-y)
- This article discusses the results of an expert panel considering treatment options after patients with advanced wild-type lung adenocarcinoma progress following immunotherapy.**
- Molife C, Hess LM, Cui ZL, et al. Sequential therapy with ramucirumab and/or checkpoint inhibitors for non-small cell lung cancer in routine practice. *Future Oncol*. 2019;15(25):2915–2931. doi: [10.2217/fon-2018-0876](https://doi.org/10.2217/fon-2018-0876)
- This article compares overall survival outcomes of patients treated with single-agent or combination ramucirumab vs. immune checkpoint inhibitor treatment.**
- Corral J, Majem M, Rodríguez-Abreu D, et al. Efficacy of nintedanib and docetaxel in patients with advanced lung adenocarcinoma treated with first-line chemotherapy and second-line immunotherapy in the nintedanib NPU program. *Clin Transl Oncol*. 2019;21(9):1270–1279. doi: [10.1007/s12094-019-02053-7](https://doi.org/10.1007/s12094-019-02053-7)
- Shiono A, Kaira K, Mouri A, et al. Improved efficacy of ramucirumab plus docetaxel after nivolumab failure in previously treated non-small cell lung cancer patients. *Thorac Cancer*. 2019;10(4):775–781. doi: [10.1111/1759-7714.12998](https://doi.org/10.1111/1759-7714.12998)
- Abu-Sbeih H, Ali FS, Naqash AR, et al. Resumption of immune checkpoint inhibitor therapy after immune-mediated colitis. *J Clin Oncol*. 2019;37(30):2738–2745. doi: [10.1200/JCO.19.00320](https://doi.org/10.1200/JCO.19.00320)
- Brahmer JR, Rodríguez-Abreu D, Robinson AG, et al. LBA51 KEYNOTE-024 5-year OS update: first-line (1L) pembrolizumab (pembro) vs platinum-based chemotherapy (chemo) in patients (pts) with metastatic NSCLC and PD-L1 tumour proportion score (TPS) ≥50%. *Ann Oncol*. 2020;31:S1181–S1182. doi: [10.1016/j.annonc.2020.08.2284](https://doi.org/10.1016/j.annonc.2020.08.2284)
- Feinberg BA, Zettler ME, Klink AJ, et al. Comparison of solid tumor treatment response observed in clinical practice with response reported in clinical trials. *JAMA Netw Open*. 2021;4(2):e2036741. doi: [10.1001/jamanetworkopen.2020.36741](https://doi.org/10.1001/jamanetworkopen.2020.36741)
- Eisenhauer EA, Therasse P, Bogaerts J, et al. New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1). *Eur J Cancer*. 2009;45(2):228–247. doi: [10.1016/j.ejca.2008.10.026](https://doi.org/10.1016/j.ejca.2008.10.026)
- Friedlaender A, Liu SV, Addeo A. Tracking the tail. *J Immuno Ther Cancer*. 2020;8(2):e000971. doi: [10.1136/jitc-2020-000971](https://doi.org/10.1136/jitc-2020-000971)
- Moliner L, Spurgeon L, Califano R. Controversies in NSCLC: which second-line strategy after chemo-immunotherapy? *ESMO Open*. 2023;8(2):100879. doi: [10.1016/j.esmoop.2023.100879](https://doi.org/10.1016/j.esmoop.2023.100879)
- Reck M, Kaiser R, Mellemegaard A, et al. Docetaxel plus nintedanib versus docetaxel plus placebo in patients with previously treated non-small cell lung cancer (LUME-Lung 1): a phase 3, doubleblind, randomised controlled trial. *Lancet Oncol*. 2014;15(2):143–155. doi: [10.1016/S1470-2045\(13\)70586-2](https://doi.org/10.1016/S1470-2045(13)70586-2)
- Garon EB, Ciuleanu TE, Arrieta O, et al. Ramucirumab plus docetaxel versus placebo plus docetaxel for second-line treatment of stage IV non-small cell lung cancer after disease progression on platinum-based therapy (REVEL): a multicentre, double-blind, randomised phase 3 trial. *Lancet*. 2014;384(9944):665–673. doi: [10.1016/S0140-6736\(14\)60845-X](https://doi.org/10.1016/S0140-6736(14)60845-X)
- Mok TS, Loong HH. A REVELation in non-small cell lung cancer treatment? *Lancet*. 2014;384(9944):640–642. doi: [10.1016/S0140-6736\(14\)60895-3](https://doi.org/10.1016/S0140-6736(14)60895-3)
- Hanna N, Shepherd FA, Fossella FV, et al. Randomized phase III trial of pemetrexed versus docetaxel in patients with non-small-cell lung cancer Previously treated with chemotherapy. *J Clin Oncol*. 2004;22(9):1589–1597. doi: [10.1200/JCO.2004.08.163](https://doi.org/10.1200/JCO.2004.08.163)
- Auclin E, Benitez-Montanez J, Tagliamento M, et al. Second-line treatment outcomes after progression from first-line chemotherapy plus immunotherapy in patients with advanced nonsmall cell lung cancer. *Lung cancer. Amsterdam (Netherlands): Elsevier*; 2023. p. 178 116–122.
- Franchi M, Pellegrini G, Corrao G. Effectiveness and cost-effectiveness profile of second-line treatments with nivolumab, Pembrolizumab, and Atezolizumab in patients with advanced NonSmall cell lung cancer. *Pharmaceuticals*. 2022;15(4):489. doi: [10.3390/ph15040489](https://doi.org/10.3390/ph15040489)
- Niki M, Nakaya A, Kurata T, et al. Immune checkpoint inhibitor re-challenge in patients with advanced non-small cell lung cancer. *Oncotarget*. 2018;9(64):32298–32304. doi: [10.18632/oncotarget.25949](https://doi.org/10.18632/oncotarget.25949)
- Giaj Levra M, Cotté FE, Corre R, et al. Immunotherapy rechallenge after nivolumab treatment in advanced non-small cell lung cancer in the real-world setting: a national data base analysis. *Lung cancer. Amsterdam (Netherlands): Elsevier*; 2020. p. 140 99–106.